TOPIC 2 · PERSONALISATION, .RULES AND SKILLS FOR EXCEL
LAB 5 THEME · CUSTOM PARETO CHART SKILL

Lab 5 — Find the Vital Few Failure Modes with a Pareto Skill

Maps to	LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence.
Duration	30 minutes
Tools	Excel Copilot · Microsoft 365 Copilot web · OneDrive (Skills folder only)
Mock data	data/Northstar_Work_Orders.xlsx
Lab files	data/templates/water-service-pareto/SKILL.md (Reviewed Pareto Skill fallback)
Course	GenAI for Engineers · v2.4
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-5

Goal

Engineers use Pareto charts to find the vital few failure modes behind most of the work. Make a chart with Excel's built-in @chart-design skill, then install a custom Pareto skill that sorts failure modes from highest to lowest.

What You Will Build

A ChartDesign_Demo sheet, a Pareto_Failure_Mode chart for all work orders and one for the overdue jobs.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac. Download this lab folder and open the files straight from its data folder; save any new file in the same folder. All data is fictional.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Turn on the built-in chart skill. Open data/Northstar_Work_Orders.xlsx in Excel. In Copilot select Settings (...) > Manage plugins & skills and turn on chart-design under Visualization.

PROMPT / CHECK — copy and adapt

```
Excel : Copilot > Settings (...) > Manage plugins & skills
Turn on : chart-design (under Visualization)
```

Step 2. Use it. Choose Allow editing and send the prompt. Select Done to keep the change, or Undo to reverse it. The bars run from highest to lowest, starting with Corrosion.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@chart-design Make a bar chart of the number of work orders for each Failure_Mode on a new
sheet named ChartDesign_Demo, sorted from highest to lowest.
```

Step 3. Upload the skill to OneDrive. In Excel open Copilot and select Settings (...) > Manage plugins & skills > Custom skills. Turn Custom skills on and select Add in OneDrive (or Create OneDrive folder, then Open skills folder). OneDrive opens at your skills folder. Select Create or upload > Folder upload and choose the water-service-pareto folder in this lab's data/templates folder.

PROMPT / CHECK — copy and adapt

```
OneDrive : Documents > Copilot > Microsoft Excel > skills > water-service-pareto > SKILL.md
```

Step 4. Turn the skill on. Back in Excel select Settings (...) > Manage plugins & skills > Custom skills > Refresh, then turn water-service-pareto on. Close the dialog and type @wat in the Copilot prompt box: water-service-pareto should appear. If not, wait a minute and select Refresh again.

PROMPT / CHECK — copy and adapt

```
Check : water-service-pareto appears when you type @
```

Step 5. Find the vital few. Select New chat, choose Allow editing, send the prompt and select Done. You should see Corrosion 42, Joint seal leak 30, Valve seat wear 22, Pump bearing wear 10, Pressure surge 8, Meter fault 4, Blockage 2, Third-party damage 2. The top 3 failure modes cause 78% of all work orders.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@water-service-pareto Make a Pareto chart of Failure_Mode for all work orders in
tblWorkOrders.
```

Step 6. Look at the overdue jobs. Send the prompt and select Done. Among the 50 overdue jobs the order changes: Joint seal leak 14, Corrosion 13, Valve seat wear 8, Pump bearing wear 7, Pressure surge 3, Blockage 2, Meter fault 2, Third-party damage 1. A Pareto shows where to look first; confirm the root cause on site.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@water-service-pareto Now make the same Pareto chart for the overdue work orders only.
```

Copy buttons for every prompt: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-5

Test It

Check	Expected result
Built-in skill	chart-design turned on; ChartDesign_Demo sorted highest to lowest.
All work orders	Corrosion 42, Joint seal leak 30, Valve seat wear 22, Pump bearing wear 10, Pressure surge 8, Meter fault 4, Blockage 2, Third-party damage 2; top 3 = 78%.
Overdue only	Joint seal leak 14, Corrosion 13, Valve seat wear 8, Pump bearing wear 7, Pressure surge 3, Blockage 2, Meter fault 2, Third-party damage 1.

Troubleshooting

If this happens	Fix
Copilot cannot see my file	In Excel, save the file and open Copilot again. In Word, PowerPoint or Outlook, upload the file with + > Upload images and files, and upload it again after you change it.
A number is different from the lab	Clear any slicers or filters, check the table has no extra rows, then ask again in a new chat.
A button or feature is missing	Check you are signed in with your course account and Microsoft 365 is up to date, then tell your trainer.

Challenge

Ask Microsoft 365 Copilot on the web to write your own Pareto SKILL.md and compare it with the supplied one.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?